

# Nicholas A. Scoville

## Curriculum Vitae

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101K Pfahler  
Ursinus College  
Collegeville, PA

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E-Mail: [nscoville@ursinus.edu](mailto:nscoville@ursinus.edu)  
URL: <https://nscoville.github.io/>

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### POSITIONS HELD

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#### Ursinus College

|                                           |                        |
|-------------------------------------------|------------------------|
| Full Professor                            | fall 2021- present     |
| Joseph Beardwood III Chair of Mathematics | fall 2017-present      |
| Associate Professor                       | fall 2016- spring 2021 |
| Chair of Mathematics and Computer Science | spring 2016-present    |
| Assistant Professor                       | fall 2010-spring 2016  |

#### Faulkner University

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|-------------------|------|
| Adjunct Professor | 2015 |
|-------------------|------|

#### Regina Luminis Academy

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| Adjunct Instructor | 2020-present |
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### EDUCATION

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#### Dartmouth College

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|------------------------------|-----------|
| Ph.D., Mathematics           | June 2010 |
| Masters of Arts, Mathematics | June 2007 |

#### Western Michigan University

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| Masters of Science, Mathematics   | June 2005   |
| Bachelors of Science, Mathematics | August 2003 |

#### Grand Rapids Community College

|                                    |          |
|------------------------------------|----------|
| Associates, Architectural Drafting | May 2001 |
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### PUBLICATIONS

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Bastian Rieck, Nicholas A. Scoville, “Expressivity Beyond Isomorphism: To Boldly Go Where Weisfeiler and Leman Did Not Go Before,” *in preparation*

Pawel Dlotko, Julian Brüggemann, Nicholas A. Scoville, “The sequential star matching on the Vietoris-Rips complex,” *in preparation*

Julian Brüggemann, David Mosquera-Lois, Nicholas A. Scoville, “A Lusternik–Schnirelmann theorem using collapsible covers,” *in preparation*

Nicholas A. Scoville, “Generalized gradient vector fields and the complex of discrete Morse intervals,” *submitted*

Nicholas A. Scoville, “The complex of discrete Morse matchings of the  $n$ -simplex: homotopy types and structural results,” *submitted*

Gregory Lupton, Nicholas A. Scoville, P. Christopher Staecker, “The Face Group of a Simplicial Complex,” *submitted*

Dominic Klyve and Nicholas A. Scoville, “Summation graphs and discrete Morse theory,” *submitted*

Nikola Milićević, Nicholas A. Scoville, “A McCord-type theorem for pseudotopological spaces and directed graphs,” *Journal of Applied and Computational Topology*, 10, 14 (2026). <https://doi.org/10.1007/s41468-026-00246-y>

Kevin P. Knudson, Nicholas A. Scoville, “Discrete Morse theory for open complexes,” *Topology and its Applications*, Volume 388, 2026, 109867, ISSN 0166-8641, <https://doi.org/10.1016/j.topol.2026.109867>

Julian Brüggemann and Nicholas A. Scoville, “On cycles and merge trees,” *Journal of Pure and Applied Algebra*, Volume 229, Issue 7, July 2025, 107967

Gregory Lupton, Oleg Musin, Nicholas A. Scoville, P. Christopher Staecker, Jonathan Treviño-Marroquin, “A Second Homotopy Group for Digital Images,” *Journal of Algebraic Combinatorics* (2024). <https://doi.org/10.1007/s10801-024-01352-9>

Connor Donovan and Nicholas A. Scoville, “Star clusters in the Matching, Morse, and Generalized complex of discrete Morse functions,” *New York J. Math.*, 29 (2023) 13931412.

Nicholas A. Scoville, “The Closure Operation as the Foundation of Topology: A Mini-Primary Source Project for Topology Students,” *Convergence* (June 2023)

Gregory Lupton, John Oprea, and Nicholas A. Scoville, “The digital Hopf construction,” *Topology and its applications*, 2023, 108405, ISSN 0166-8641, <https://doi.org/10.1016/j.topol.2022.108405>.

Benjamin Johnson and Nicholas A. Scoville, “Merge trees in discrete Morse theory,” *Research in the Mathematical Sciences* 9, 49 (2022). <https://doi.org/10.1007/s40687-022-00347-x>

Gregory Lupton and Nicholas A. Scoville, “Digital Fundamental Groups and Edge Groups of Clique Complexes,” *Journal of Applied and Computational Topology*, (2022). <https://doi.org/10.1007/s41468-022-00095-5>

Desamparados Fernandez-Ternero, Enrique Macias-Virgos, David Mosquera-Lois, Nicholas A. Scoville, and Jose-Antonio Vilches, “Fundamental Theorems of Morse theory on posets,” *AIMS Mathematics* 2022, Volume 7, Issue 8: 14922-14945. doi: 10.3934/math.2022818

Gregory Lupton, John Oprea, and Nicholas A. Scoville, “Subdivision of Maps of Digital Images,” *Discrete and Computational Geometry*, 67, 698742 (2022). <https://doi.org/10.1007/s00454-021-00350-z>

Nicholas A. Scoville and Matthew C. B. Zaremsky, “Higher connectivity of the Morse complex,” *Proceedings of the AMS Series B*, 9 (2022), 135149.

Connor Donovan, Maxwell Lin, and Nicholas A. Scoville, “On the homotopy and strong homotopy type of complexes of discrete Morse functions,” *Canadian Mathematical Bulletin*, 1-19, 2022, doi:10.4153/S0008439522000121

Gregory Lupton, John Oprea, and Nicholas A. Scoville, “Homotopy Theory in Digital Topology,” *Discrete and Computational Geometry*, 67 (2022), no. 1, 112165, doi.org/10.1007/s00454-021-00336-x

Maxwell Lin and Nicholas A. Scoville, “On the automorphism group of the Morse complex,” *Advances in Applied Mathematics*, Volume 131, October 2021, 102250

Gregory Lupton, John Oprea, and Nicholas A. Scoville, “A Fundamental Group for Digital Images,” *Journal of Applied and Computational Topology*, 5 (2021), no. 2, 249311.

Nicholas A. Scoville, “Topology from Analysis: A Mini-Primary Source Project for Topology Students,” *Convergence* (June 2020)

Desamparados Fernandez-Ternero, Enrique Macias-Virgos, Nicholas A. Scoville, and Jose-Antonio Vilches, “Strong discrete Morse theory and simplicial LusternikSchnirelmann category: A discrete version of the Lusternik-Schnirelmann Theorem,” *Discrete and Computational Geometry*, 63 (2020), no. 3, 607623.

Ian Rand and Nicholas A. Scoville, “Discrete Morse functions, vector fields, and homological sequences on trees,” *Involve, A Journal of Mathematics* Involve, a Journal of Mathematics 13-2 (2020), 219–229. DOI 10.2140/involve.2020.13.219

Yuqing Liu and Nicholas A. Scoville, “The realization problem for discrete Morse functions on trees,” *Algebra Colloquium*, **27** : 3 (2020) 455–468 DOI: 10.1142/S1005386720000371

Nicholas A. Scoville, “The Cantor Set Before Cantor: A Mini-Primary Source Project for Analysis and Topology Students,” *Convergence* (May 2019)

Mike Agiorgousis, Brian Green, Alex Onderdonk, Nicholas A. Scoville, and Kim Rich, “Homological sequences in discrete Morse theory,” *Topology Proceedings*, 54 (2019) 283–294

Colin Adams, Allison Henrich, Kate Kearney and Nicholas A. Scoville, “Knots Related by Knotoids,” *American Mathematical Monthly* Volume 126, 2019 - Issue 6, 483–490

Nicholas A. Scoville and Karthik Yegnesh “A Persistent Homological Analysis of Network Data Flow Malfunctions,” *Journal of Complex Networks*, Issue 6, 1 December 2017, Pages 884-892

Nicholas A. Scoville, “Connecting Connectedness: A Mini-Primary Source Project for Topology Students,” *Convergence* (October 2017)

Nicholas A. Scoville and Willie Swei “On the Lusternik–Schnirelmann category of a simplicial map,” *Topology and its applications* 216 (2017), 116-128

Brian Green, Nicholas A. Scoville, and Mimi Tsuruga, “Estimating the discrete Lusternik–Schnirelmann category,” *Topological Methods in Nonlinear Analysis*, 45, No. 1 (2015), 103–116

Akshaye Dhawan, Michelle Tanco, and Nicholas A. Scoville, “A Distributed Greedy Algorithm for Constructing Connected Dominating Sets in Wireless Sensor Networks,” *SENSORNETS*, Lisbon, Portugal January 2014

Nicholas A. Scoville, “Metric Structures for CW Complexes,” *Topology Proceedings*, 44 (2014) 117–131

Seth Aaronson, Marie Meyer, Nicholas A. Scoville, Mitchell T. Smith, and Laura Stibich, “Graph Isomorphisms in discrete Morse theory,” *AKCE Int. J. Graphs Comb.*, 11, No. 2 (2014), 163–176

Seth Aaronson and Nicholas A. Scoville, “Lusternik–Schnirelmann category for cell complexes,” *Illinois J. of Mathematics*, 57, No. 3 (2013), 743–753

Nicholas A. Scoville, “Georg Cantor at the Dawn of Point-Set Topology,” *Loci*, (March 2012), DOI: 10.4169/loci003861

Nicholas A. Scoville, “Lusternik–Schnirelmann Category and the Connectivity of  $X$ ,” *Algebraic & Geometric Topology*, 12 (2012) 435-448

Nicholas A. Scoville, “Mapping Cone Sequences and a Generalized Notion of Cone Length,” *JP Journal of Geo. and Top.*, 11(2011), Issue 3, 209-233

Nicholas A. Scoville, “A Metric for Homotopy Types,” Ph.D. Thesis, Dartmouth College, Spring 2010

Rob Nendorf, Nicholas A. Scoville, Jeff Strom, “Categorical Sequences,” *Algebraic & Geometric Topology*, 6 (2006) 809–838

## BOOKS

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*Abstract Algebra or There and Back Again: A Polynomial's Tale*, Princeton University Press  
(forthcoming)

*Teaching and Learning with Primary Source Projects*, editor with Janet Barnett and David K. Ruch, AMS/MAA Press, 2023

*Discrete Morse theory*, AMS Press, 2019

Student Mathematical Library series

## BOOK CHAPTERS

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Nicholas A. Scoville, “Sometimes when your hopes have all been shattered,” *Living Proof: Stories of resilience along the mathematical journey*, Edited by Henrich et al., AMS/MAA Press, 2019

## ARTICLES

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Nicholas A. Scoville, “Course on Mathematics for Human Flourishing” *MAA FOCUS* February/March 2023

## BOOK REVIEWS

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Nicholas A. Scoville, “A Brief Quadrivium” by Peter Ulrickson, *The American Mathematical Monthly* Volume 131, 2024 - Issue 6

Nicholas A. Scoville, “Never a dull moment: Hassler Whitney, Mathematics Pioneer” by Keith Kendig, *The American Mathematical Monthly* Volume 126, 2019 - Issue 9

## HONORS AND AWARDS

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Collaborative Research: AF: New Frontiers of Discrete Morse Theory for Scientific Data Analysis and Machine Learning

NSF Future Core

with Henry Adams, Aurora Clark,

Kevin Knudson, and Bei Wang

(DMS-2617597, submitted)

AMS-Simons Research Enhancement Grants for PUI Faculty (\$9,000)

July 2023-June 2026

Laughlin Professional Achievement Award

for distinguished service to Ursinus college through significant contribution to scholarship

May 2023

Paul R. Halmos-Lester R. Ford Award

for article of expository excellence published in The American Mathematical Monthly

August 2020

REU SITE: Exploration and Professional

Excellence in the Mathematical Sciences

NSF Grant 1851948 (\$225,469), Principal Investigator

April 2021- March 2023

|                                                                                                                                                                                                                                                     |                       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| Western Michigan University<br>Department of Mathematics Alumni Achievement Award<br>For contributions to mathematics education and research                                                                                                        | October 2018          |
| Collaborative Research: RUI: Transforming Instruction<br>in Undergraduate Mathematics via Primary History Sources<br>NSF IUSE Grant 1524065 (\$71,002)<br>PIs at Colorado State, Central Washington, NMSU, Xavier,<br>U Colorado, Denver, U Florida | Aug. 2015- Sept. 2020 |
| Best oral presentation at HTCA conference in Genoa, Italy<br>sponsored by Gruppo Italiano Ricercatori in Pattern Recognition.                                                                                                                       | February 2015         |
| Mellon travel grant                                                                                                                                                                                                                                 | July 2013             |
| Mellon travel grant                                                                                                                                                                                                                                 | May 2012              |
| Project NExT Fellow                                                                                                                                                                                                                                 | Aug. 2010 – Aug. 2011 |

## GRADUATE STUDENT MENTORING

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|                                                                                                                                                                                                             |               |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| PhD thesis committee for Darryl Jent<br>“Reduced Product Type Monoid-Module Extension”<br>Western Michigan University,                                                                                      | May 2026      |
| Served on PhD thesis committee for David Mosquera Lois<br>“Morse Theory on Finite Spaces”<br>University of Santiago de Compostela,                                                                          | February 2022 |
| Served on MA math thesis committee for Marwa Mosallam<br>“On cup-products of cofibers of maps between Moore spaces,<br>Hopf invariant, and Lusternik–Schnirelmann category”<br>Western Michigan University, | July 2021     |
| Served on PhD thesis committee for Francisco Belchi Guillamon<br>“ $A_\infty$ -persistence”<br>University of Malaga,                                                                                        | December 2015 |

## TEACHING EXPERIENCE

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Ursinus College,

*Math 400: Mathematics for Human Flourishing*  
*Math 335: Abstract Algebra*  
*Math 491: Algebraic Topology*  
*Math 451: Discrete Morse Theory*  
*Math 361: Graph Theory*  
*CIE 100: Common Intellectual Experience*  
*Math 421: Topology*  
*Math 211: Multivariable calculus*  
*Math 235: Linear Algebra*  
*Math 10: Problem Solving*  
*Math 322: Geometry*  
*Math 341: Probability*  
*Math343: Math for Data*  
*Math 236W: Discrete Mathematics*  
*Stat 141Q: Statistics I*  
*Math 111: Calculus I*  
*Math 112: Calculus II*

**Adjunct instructor,** Regina Luminis Academy

*Aristotelian Logic*  
*Philosophy of nature*  
*Calculus*  
*Scholastic metaphysics*  
*Medieval philosophy*  
*Physics*  
*Quadrivium*

**Adjunct Professor,** Faulkner University

*Math 1312: Mathematics*

Spring 2015

**Other teaching experience,**

*Teaching Assistant: Dartmouth College*  
*Instructor: Dartmouth College*  
*Teaching Assistant: Dartmouth College*  
*Instructor: Western Michigan University*  
*Undergraduate Teaching Assistant: Western Michigan University*

Summer 2009  
 2007-2009  
 2005-2007  
 2003-2005  
 2002-2003

## INDEPENDENT STUDIES

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**Ursinus College**

Topological data analysis for biological data, Joshua Funk  
 Metrics on the Morse complex, Nathaniel Nerone  
 Graph theory, Joshua Brown  
 Category theory, Tony Delgado  
 Star clusters and homotopy type, Connor Donovan  
 Cohomology, Connor Donovan  
 Category theory, Tony Delgado  
 Discrete Morse theory and knots, Connor Donovan  
 Discrete Morse theory, Elvi Sopiqoti  
 Discrete plates and Olives, Matthew Furgele  
 Analyzing heart data, Ben Minardi

Fall 2026  
 Spring 2026  
 Spring 2026  
 Spring 2023  
 Fall 2022  
 Fall 2022  
 Fall 2022  
 Spring 2021  
 Spring 2020  
 Spring 2020  
 Spring 2019

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|---------------------------------------------------------------------|-----------------------------------------------|
| Algebraic topology, Nick Tulio                                      | Spring 2019                                   |
| Strong collapsibility and the Morse complex, Max Lin                | Spring 2019                                   |
| Group of strong self-homotopy equivalences, Vince Sergi, Ryan Quick | Spring 2019                                   |
| Boolean functions, Ryan Quick                                       | Fall 2018                                     |
| Algebraic topology, Jason Bennett                                   | Fall 2018                                     |
| Generating Discrete Morse Functions from Point Data, Ashlyn Welch   | Spring 2018                                   |
| Random discrete Morse theory, Nikolai Peralta                       | Spring 2018                                   |
| Homotopy theory, Karthik Yegnesh                                    | Fall 2016 Spring 2017, Fall 2017, Spring 2018 |
| Topology, Ian Rand                                                  | Spring 2017                                   |
| Discrete Morse theory and Persistent homology, Yuqing Liu           | Spring 2017                                   |
| Random discrete Morse theory, Nikolai Peralta, Chase Babrich        | Fall 2016, Spring 2017                        |
| Algebraic topology, Matan Peleg                                     | Fall 2016, Spring 2017                        |
| Category theory, Michael Vennettilli                                | Spring 2015                                   |
| Multiplication in discrete Morse theory, Rose Blanchard             | Spring 2015                                   |
| Number theoretic notions in discrete Morse theory, Ian Rand         | Spring 2015                                   |
| Discrete Morse theory, Tyler Helms                                  | Fall 2014                                     |
| Estimating the discrete LS category, Brian Green                    | Fall 2013                                     |
| Homology and Cohomology, Seth Aaronson, Brian Green, Michelle Tanco | Spring 2013                                   |
| Discrete LS category, Seth Aaronson                                 | Spring 2013                                   |
| Counting discrete Morse functions, Seth Aaronson                    | Spring 2012                                   |
| Lie Groups, Brian Green                                             | Spring 2012                                   |
| Discrete Morse Theory, Mike Agiorgousis                             | Spring 2012                                   |
| Markov Chains, Jayant Velagala                                      | Spring 2012                                   |
| Probability, Jayant Velagala                                        | Fall 2011                                     |
| Knot Theory, Will Molden                                            | Fall 2011                                     |
| Discrete Morse Theory, Seth Aaronson                                | Spring 2011                                   |

## STUDENT POSTERS AND PRESENTATIONS

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| Jose Arbelo, Tony Delgado, Charley Kirk, Zach Schlamowitz “ <i>The Dope metric is SIC: A stable, informative, and computable metric on time series</i> ,”<br>Mathfest, Philadelphia, Pennsylvania                | August 2022  |
| Connor Donovan “ <i>Towards the homotopy type of the Morse complex</i> ,”<br>Exploring Innovation in Appalachia, (virtual), won third place in the<br>math/physics/astronomy category,                           | August 2021  |
| Connor Donovan “ <i>Towards the homotopy type of the Morse complex</i> ,”<br>Mathfest, (virtual)                                                                                                                 | August 2021  |
| Benjamin Johnson “ <i>Merge trees in discrete Morse theory</i> ,”<br>Mathfest, Cincinnati, Ohio                                                                                                                  | August 2019  |
| Yuqing Liu “ <i>Persistence equivalence of discrete Morse functions on trees</i> ,”<br>Mathfest, Denver, Colorado                                                                                                | August 2018  |
| Karthik Yegnesh “ <i>Families of Objects in Categories and Elementary Topoi</i> ,”<br>AMS/MAA Joint Math Meetings, Atlanta, Georgia                                                                              | January 2017 |
| Karthik Yegnesh “ <i>Cosheaf theoretical constructions in networks and persistent homology</i> ,”<br>68 <sup>th</sup> annual Delaware Valley Science Fair, Oaks, Pennsylvania (won “first place” in Mathematics) | March 2016   |

- Karthik Yegnesh "*Cosheaf theoretical constructions in networks and persistent homology*,"  
59<sup>th</sup> annual Montgomery County Science Research Competition, Collegeville, Pennsylvania  
(won "first place" in math category) March 2016
- Matt Belle "*Arboricity*,"  
AMS/MAA Joint Math Meetings, Baltimore, Maryland January 2014
- Brian Green "*Estimating the discrete Lusternik–Schnirelmann category*,"  
AMS/MAA Joint Math Meetings, Baltimore, Maryland January 2014
- Seth Aaronson "*Lusternik–Schnirelmann category for cell complexes*,"  
AMS/MAA Joint Math Meetings, San Diego, California January 2013
- Mike Agiorgousis, Brian Green, and Alex Onderdonk "*Discrete Morse Functions and Homology*,"  
AMS/MAA Joint Math Meetings, San Diego, California (won "Outstanding Presentation" award) January 2013
- Mike Agiorgousis, Brian Green, and Alex Onderdonk "*Discrete Morse Functions and Homology*,"  
Undergraduate Science Research Symposium, Haverford College September 2012
- Mike Agiorgousis, Brian Green, and Alex Onderdonk "*Discrete Morse Functions and Homology*,"  
Mathfest, Madison, Wisconsin August 2012
- Mike Agiorgousis, Brian Green, Alex Onderdonk, and Kim Rich "*Discrete Morse Functions and Homology*,"  
*Disappearing Boundaries Summer Research Meeting*, Lebanon Valley College July 2012
- Seth Aaronson and Marie Meyer, "*Graph Isomorphisms in Discrete Morse Theory*," *AMS/MAA Joint Meetings*,  
Boston, MA January 2012

## INVITED PRESENTATIONS (past 10 years) —————

- Red Light, Green Light: How AI Text Quietly Betrays Its Own Fingerprint* October, 2026  
Ursinus MAA student club talk
- The complex of discrete Morse matchings as the space of tie-breakings* October, 2026  
Advances in Applied Topology, AMS Eastern  
Sectional Meeting, George Washington University
- Discrete Morse Theory for Open Complexes (plenary lecture)* July, 2026  
II Conference on Algebraic Topology and Related Topics, University of Santiago de Compostela
- Perfect discrete Morse matchings on VR complexes: a preliminary report* March, 2026  
TDA: theory and applications, AMS Southeastern  
Sectional Meeting, Southern Georgia University Armstrong campus
- A McCord theorem for Čech closure spaces* February, 2026  
Dioscuri Centre in TDA Basic Notions and Applied Topology seminar, Warsaw Poland
- Discrete Morse theory for open simplicial complexes* January, 2026  
AMS/MAA Joint Meetings, Washington DC
- Discrete Morse theory as an introduction to topology* October, 2025  
The College of New Jersey undergraduate seminar
- Discrete Morse theory: Introduction and research directions* July, 2025  
AATRC Topological Complexity Virtual Seminar

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|------------------------------------------------------------------------------------------------------|----------------|
| <i>Discrete Morse theory for open simplicial complexes</i>                                           | July, 2024     |
| AMS-UMI Joint Math meetings, Palermo University                                                      |                |
| <i>Discrete Morse theory as an introduction to topology</i>                                          | April, 2024    |
| Towson University undergraduate conference, Plenary talk                                             |                |
| <i>Tools of discrete Morse theory</i>                                                                | February, 2024 |
| Applied and Combinatorial topology workshop, Dagstuhl Germany                                        |                |
| <i>A McCord theorem for closure spaces</i>                                                           | October, 2023  |
| Topology seminar, Notre Dame University                                                              |                |
| <i>The Moore Method, Flatland, and alternate topology axioms</i>                                     | July, 2023     |
| Innovative Pedagogy in Geometry and Topology, Oberlin College                                        |                |
| <i>Discrete Morse theory as an introduction to topology</i>                                          | July, 2023     |
| Euler Circle                                                                                         |                |
| <i>Discrete Morse theory as an introduction to topology</i>                                          | April, 2023    |
| University of Arizona undergraduate seminar                                                          |                |
| <i>Towards a new digital homotopy theory</i>                                                         | March, 2023    |
| Discrete homotopy workshop, American Institute of Mathematics, San Jose                              |                |
| <i>Discrete Morse theory: three approaches</i>                                                       | October, 2022  |
| Michael Penn math Patreon Seminar                                                                    |                |
| <i>Some recent results on the homotopy type of the Morse complex (semi-plenary lecture)</i>          | July, 2022     |
| Applied Topology in Bedlewo, Bedlwo Poland                                                           |                |
| <i>The homotopy type of the Morse complex for some collections of trees</i>                          | April, 2022    |
| Virtual AMS/MAA Joint Meetings                                                                       |                |
| <i>Towards a new digital homotopy theory</i>                                                         | December, 2021 |
| Virtual University of Regina seminar                                                                 |                |
| <i>Discrete Morse theory as an introduction to topology</i>                                          | November, 2021 |
| Virtual Rutgers undergraduate seminar                                                                |                |
| <i>Higher connectivity of the Morse complex</i>                                                      | January, 2021  |
| Virtual AMS/MAA Joint Meetings                                                                       |                |
| <i>Discrete Morse theory as an introduction to topology</i>                                          | March, 2020    |
| Juniata College math colloquium                                                                      |                |
| <i>Towards a new digital homotopy theory</i>                                                         | February, 2020 |
| University of Albany Geometry/Topology seminar                                                       |                |
| <i>Digital topology: A smooth introduction</i>                                                       | November, 2019 |
| Westminster College, Fulton Missouri                                                                 |                |
| <i>Towards a new digital homotopy theory</i>                                                         | November, 2019 |
| University of Missouri Geometry/Topology seminar                                                     |                |
| <i>Strong discrete Morse theory and an application to simplicial Lusternik–Schnirelmann category</i> | November, 2019 |
| Topological Complexity and Related topics, AMS Southeastern Sectional Meeting, University of Florida |                |

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| <i>Towards a new digital homotopy theory</i><br>Lehigh University Algebraic Topology seminar                                                                                                                                                                        | May, 2019       |
| <i>Digital Topology: A smooth introduction</i><br>Western Michigan University                                                                                                                                                                                       | October, 2018   |
| <i>Digital Topology: A smooth introduction</i><br>Colloquium, Elon University                                                                                                                                                                                       | March, 2018     |
| $S^1$ and $S^2$ and $S^3$ , oh fy! <i>A digital Hopf fibration</i><br>Math Colloquium, Montana State University                                                                                                                                                     | January, 2018   |
| $S^1$ and $S^2$ and $S^3$ , oh fy! <i>A digital Hopf fibration</i><br>Colloquium, Catholic University of America                                                                                                                                                    | November, 2017  |
| <i>Digital Topology: A smooth introduction</i><br>Colloquium, Bard College                                                                                                                                                                                          | November, 2017  |
| $S^1$ and $S^2$ and $S^3$ , oh fy! <i>A digital Hopf fibration</i><br>Colloquium, Dartmouth College                                                                                                                                                                 | November, 2017  |
| <i>Digital Topology: A smooth introduction</i><br>Colloquium, Seattle University                                                                                                                                                                                    | October, 2017   |
| <i>Digital Topology: A smooth introduction</i><br>Colloquium, Central Washington University                                                                                                                                                                         | October, 2017   |
| <i>Simplicial Lusternik–Schnirelmann category and strong discrete Morse theory</i><br>Topology Seminar, University of Florida                                                                                                                                       | October, 2017   |
| $S^1$ and $S^2$ and $S^3$ , oh fy! <i>A digital Hopf fibration</i><br>Colloquium, University of Florida Colloquium                                                                                                                                                  | October, 2017   |
| <i>Digital Topology: A smooth introduction</i><br>Math Club, Cleveland State University                                                                                                                                                                             | September, 2017 |
| <i>Towards a new digital homotopy theory</i><br>Colloquium, Cleveland State University                                                                                                                                                                              | April 2017      |
| <i>Collaborative Research: Transforming Instruction in Undergraduate Mathematics via Primary Historical (TRIUMPHS)</i><br>MAA Invited Paper Session on Research in Improving Undergraduate Mathematical Sciences Education Program, AMS/MAA Joint Meetings, Atlanta | January, 2017   |
| <i>(Strong) discrete Morse theory as an introduction to topology</i><br>Colloquium, Butler University                                                                                                                                                               | September, 2016 |
| <i>Georg Cantor at the dawn of point-set topology</i><br>Butler University                                                                                                                                                                                          | September, 2016 |

## WORKSHOPS RUN

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|                                               |                 |
|-----------------------------------------------|-----------------|
| <i>Open problems in discrete Morse theory</i> | September, 2023 |
|-----------------------------------------------|-----------------|

|                                                                                                            |                       |
|------------------------------------------------------------------------------------------------------------|-----------------------|
| BIGS Young Researcher Networking: Meeting on<br>Topology and Applications, Bonn Germany (invited workshop) |                       |
| <i>Discrete Morse theory: Breadth and depth</i>                                                            | July, 2022            |
| Adam Mickiewicz University, Poznan Poland (invited workshop)                                               |                       |
| <i>MAA Workshop: Teaching Undergraduate Mathematics via Primary Source Projects.</i>                       | January, 2020         |
| AMS/MAA Joint Math Meetings, Denver Colorado                                                               |                       |
| <i>TRIUMPHS Graduate student training Workshop</i>                                                         | July 19-20, 2019      |
| New Mexico State University                                                                                |                       |
| <i>TRIUMPHS Training Workshop</i>                                                                          | September 13-15, 2018 |
| University of Colorado Denver                                                                              |                       |
| <i>Teaching Undergraduate Mathematics via Primary Source Projects</i>                                      | January 2018          |
| AMS/MAA Joint Math Meetings, San Diego CA                                                                  |                       |
| <i>Teaching Mathematics with Primary Historical Sources</i>                                                | April 1, 2017         |
| MAA EPADEL sectional meeting, Kutztown PA                                                                  |                       |
| <i>TRIUMPHS Training Workshop</i>                                                                          | September 8-10, 2016  |
| University of Colorado Denver                                                                              |                       |
| <i>Connecting Past to Present: An approach to teaching topology via original resources</i>                 | July, 2016            |
| HPM 2016, Montpellier France                                                                               |                       |

## SERVICE TO URSINUS COLLEGE

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|                                                                                                                                                          |                      |
|----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| Chair, Department of Mathematics, Computer Science, and Statistics                                                                                       | Spring 2016-present  |
| Search Committee, Department of Mathematics, Computer Science, and Statistics;<br>Member every year since 2010 (Chair while serving as Department Chair) | 2010-present         |
| Academic Council; Member                                                                                                                                 | 2013-2018, 2023-2025 |
| Academic Standards and Discipline; Member                                                                                                                | 2018-2020            |
| Information Technology; Member                                                                                                                           | 2026-present         |
| Arts and Lecture Committee; Member                                                                                                                       | 2025-2026            |
| Calculus Committee, Ursinus College; Member                                                                                                              | August 2010-present  |
| Statistics Committee, Ursinus College; Member                                                                                                            | August 2010-present  |
| Kappa Mu Epsilon Faculty Advisor                                                                                                                         | 2014-present         |

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|-------------------------------------------------------------------|-----------------------|
| Faculty Advisor, Ursinus MAA Student Chapter                      | 2014-2016             |
| Organizer for Ursinus College $\epsilon$ -talks                   | Fall 2010-Fall 2016   |
| Senior Personnel, Ursinus College REU (NSF Grant No. DMS-1003972) | June 2012-August 2012 |
| Senior Personnel, Ursinus College REU (NSF Grant No. DMS-1003972) | June 2011-August 2011 |

## PROFESSIONAL SERVICE

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| Founding Editorial Board Member, Incepta Mathematica                                                                                                                  | May 2026 - present         |
| Scientific Committee, II Conference on Algebraic Topology<br>and Related Topics Universidade de Santiago de Compostela                                                | July 2026                  |
| Served on MAA panel “Designing and Running Effective Undergraduate Research Programs”<br>Joint Math Meetings, Washington DC                                           | January 2026               |
| MAA Dolciani Mathematical Expositions , editorial board                                                                                                               | January 2024-present       |
| Referee for several journals                                                                                                                                          | 2013-present               |
| Scientific Committee, Conference on Algebraic Topology<br>and Related Topics Universidade de Santiago de Compostela                                                   | April 2025                 |
| Organized Special Session “Applied and computational topology”<br>with Kate Turner, Joint Meeting of the NZMS, AustMS and AMS,<br>University of Auckland, New Zealand | December 2024              |
| Reviewer for MathSciNet Mathematical Reviews                                                                                                                          | February 2015-January 2024 |
| Reviewer for mathematical publication database Zentralblatt                                                                                                           | August 2012-January 2024   |
| Focus Magazine, editorial board                                                                                                                                       | November 2017-2022         |
| Member of NSF’s College of Reviewers for Undergraduate Education                                                                                                      | 2018-2021                  |
| Served on virtual panel “Teaching and the Liberal Arts”<br>University of Tennessee, Knoxville                                                                         | April 2020                 |

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| Served on MAA Basic Library List Committee                                                                                                                                          | January 2017-January 2020 |
| Book reviewer for online MAA book reviews                                                                                                                                           | 2014-2020                 |
| Scientific Committee, ESU8, Oslo Norway                                                                                                                                             | July 2018                 |
| Scientific Committee, ICART 2018, Rabat Morocco                                                                                                                                     | July 2018                 |
| Organized Special Session “AMS Special Session on Open & Accessible Problems for Undergraduate Research, ”<br>with Allison Henrich and Michael Dorff at AMS/MAA Joint Math Meetings | January 2018              |
| Served on NSF panel review                                                                                                                                                          | 2016, 2017, 2018          |
| CUR Councilor in the Mathematics and Computer Sciences Division                                                                                                                     | 2014-2017                 |
| Organized Special Session “AMS Special Session on Open & Accessible Problems for Undergraduate Research, ”<br>with Allison Henrich and Michael Dorff at AMS/MAA Joint Math Meetings | January 2017              |
| Served on panel “The Research and Teaching Pendulum: Finding a Stable Equilibrium” at AMS/MAA Joint Math Meetings                                                                   | January 2017              |
| Organized Special Session “Applied and Computational Topology, ”<br>with Matthew Wright and Paweł Dłotko at AMS/MAA Joint Math Meetings                                             | January 2016              |
| Organized panel “Finding a thesis topic and advisor, ” at<br>AMS/MAA Joint Math Meetings                                                                                            | January 2016              |
| Reviewed applications for Posters on the Hill                                                                                                                                       | Fall 2015                 |
| Treasurer for YMN (Young Mathematicians Network)                                                                                                                                    | 2010-2015                 |
| Organized panel “Graduate school: Choosing one, getting in, staying in, ” at<br>AMS/MAA Joint Math Meetings                                                                         | January 2015              |
| Reviewed applications for Posters on the Hill                                                                                                                                       | Fall 2014                 |
| Served on panel “You published your dissertation: now what?” at<br>AMS/MAA Joint Math Meetings                                                                                      | January 2013              |

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| Organized panel “The on-campus interview survival guide” at AMS/MAA Joint Math Meetings                        | January 2013             |
| Organized Panel “Hit the Ground Running! Interview like a Pro and land the job” at AMS/MAA Joint Math Meetings | January 2012             |
| Judge for MAA Student Paper Session 3, MathFest                                                                | August 2011              |
| Judge of research abstracts for Young Mathematicians Network Conference applicants                             | July 2011                |
| Judge for MAA Student Poster Session, Joint Mathematics Meetings                                               | January 2011             |
| Judge for MAA Student Paper Session 11, MathFest                                                               | August 2010              |
| Reader/Reviwer for “Introduction to Homotopy Theory” by Martin Arkowitz                                        | 2007-2008                |
| Student Seminar Organizer                                                                                      | 2004-2005                |
| WMU Pi Mu Epsilon Graduate Representative                                                                      | 2003-2005                |
| Grader                                                                                                         | Summer 2004, Summer 2003 |

## ADDITIONAL ACTIVITIES

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| <b><i>Equipment manager</i></b> RLA Cross Country and Track<br>Berwyn, PA                                                                                            | August 2024-present   |
| <b>Tutor for Critical Point Test Prep</b><br>Berwyn, PA                                                                                                              | Fall 2020-Fall 2022   |
| <b>Director of Religious Education</b><br>Sacred Heart Parish, Royersford PA.                                                                                        | Fall 2019-Fall 2023   |
| <b>Second Grade PREP Teacher</b><br>Taught hour-long once weekly religious formation/catechism class to second grade students at Sacred Heart Parish, Royersford PA. | Fall 2016-Spring 2019 |
| <b>First Grade PREP Teacher</b><br>Taught hour-long once weekly religious formation/catechism class to first grade students at Sacred Heart Parish, Royersford PA.   | Fall 2015-Spring 2016 |
| <b>Fourth Grade PREP Teacher</b><br>Taught hour-long once weekly religious formation/catechism class to fourth grade students at Sacred Heart Parish, Royersford PA. | Fall 2014-Spring 2015 |

**Ursinus College Newman Society Faculty Advisor**

Fall 2011-present

**Sixth Grade PREP Teacher**

Fall 2011-Spring 2014

Taught hour-long once weekly religious formation/catechism class to sixth grade students at Sacred Heart Parish, Royersford PA.

**Sixth Grade Religious Education Teacher**

Fall 2010-Spring 2011

Taught hour-long once weekly religious formation/catechism class to sixth grade students at St. Norbert Parish, Paoli PA.

**Seventh Grade Religious Education Teacher**

Fall 2009-Spring 2010

Taught hour-long once weekly religious formation/catechism class to seventh grade students at St. Denis Parish, Hanover NH.

**Youth Group Leader at St. Denis Parish**

Fall 2008-Spring 2009

Gave lectures, organized activities, participated in service projects